# AI Image Generator

## 11. Performance

Version 2.3
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Project owner: Frazile AI
Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Images, video and music on one GPU: Z-Image Turbo, SDXL fine-tunes, Qwen-Image, FLUX.1, LoRAs, LTX-Video, Wan 2.1, MusicGen and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and an Admin panel that manages models, LoRAs and downloads and shows live system statistics.

# 11. Performance

All figures were measured on the development machine (RTX 3090, 24 GB, driver 610.88, torch 2.14.1 with CUDA 12.6) and taken from generations.db and the v2.0 model sweep.

## 11.1 Generation by model (832x1216, prompt weighting on for SDXL)

| Model | Steps | CFG | Sampler | Inference | Peak VRAM |
|---|---|---|---|---|---|
| Z-Image Turbo | 8 | 0 | Flow-match Euler | 8.6 s (9.8 s at 1344x768) | 18.1 GB |
| Qwen-Image + Lightning (4-bit) | 8 | 1 (none) | Flow-match Euler, exp. shift | 13.4 s (13.3 to 14.2 s at 1344x768 and 1024x1024) | 18.0 GB |
| SDXL base 1.0 | 30 | 7 | DPM++ 2M Karras | 8.0 s | 13.1 GB |
| Juggernaut XL v9 | 30 | 5 | DPM++ 2M Karras | 8.0 s | 13.1 GB |
| RealVisXL V5.0 | 30 | 4 | DPM++ SDE Karras | 8.0 s | 13.2 GB |
| DreamShaper XL v2 Turbo | 6 | 2 | DPM++ SDE Karras | 2.0 s | 13.2 GB |
| Playground v2.5 | 40 | 3 | EDM DPM-Solver | 11.5 s | 13.2 GB |
| Animagine XL 3.1 | 28 | 6 | Euler a | 7.3 s | 13.2 GB |

Every SDXL-family model runs at about 3.7 to 3.8 steps per second at this size. Inference time is therefore proportional to the step count, which is why the turbo model's 6 steps finish in 2 seconds. Z-Image Turbo runs at about 1.1 s per step but needs only 8, so a finished image takes 9 to 10 s with better prompt adherence and legible text. Loading an SDXL-family model from the local cache takes 26 to 27 s; freeing it returns resident VRAM to 0.01 GB. Measurements taken while another process holds GPU memory are not valid: with the web app resident alongside the test, Z-Image took 78 to 128 s per image because Windows paged VRAM.

## 11.2 Generation (v1.0 measurements, 1024x1024 and 16:9, SDXL base)

| Size | Steps | Inference | Steps/s | Peak VRAM | Temp after | Power after |
|---|---|---|---|---|---|---|
| 1024x1024 | 15 | 4.7 to 4.9 s | 3.1 to 3.2 | 11.5 GB | 55 to 56 C | 377 to 379 W |
| 1024x1024 | 30 | 9.7 s | 3.1 | 11.5 GB | - | - |
| 1344x768 | 40 | 12.8 to 13.5 s | 3.0 to 3.1 | 11.4 GB | 52 to 54 C | 349 to 357 W |
| 768x1344 | 40 | 11.2 to 11.4 s | 3.5 to 3.6 | 11.5 GB | 55 to 57 C | 382 to 385 W |

## 11.3 Upscaling

| From | To | Passes | Time | Output file |
|---|---|---|---|---|
| 1344x768 | 3840x2194 (4K) | 1 | 3.4 s | 9.9 MB |
| 768x1344 | 2194x3840 (4K) | 1 | 2.9 s | 7.9 MB |
| 768x1344 | 1463x2560 (2K, Upscale bar) | 1 | 3.6 s | 3.5 MB |
| 768x1344 | 4388x7680 (8K) | 2 | 8.4 to 8.5 s | 30.5 MB |

## 11.4 Memory

| Stage | VRAM |
|---|---|
| Idle with one SDXL-family model loaded | 6.6 GB allocated |
| Idle with Z-Image Turbo loaded (bf16 transformer + 8-bit text encoder) | 15.8 GB allocated |

| Stage | VRAM |
|---|---|
| Idle with Qwen-Image loaded (NF4 transformer + 4-bit text encoder + Lightning LoRA) | 17.5 GB allocated |
| Peak during Qwen-Image generation at 1 MP (VAE tiling on) | 18.0 GB; 21.7 GB before VAE tiling, which caused paging |
| Peak during Z-Image generation at 1344x768 | 18.1 GB |
| Peak during 1 MP generation, plain prompt | 11.4 to 10.5 GB |
| Peak during 832x1216 generation with compel embeddings | 13.1 to 13.2 GB |
| After unload | 0.01 GB |
| Upscaler weights | 67 MB (half precision) |
| Peak during tiled upscale | no measurable increase over the generation peak |

## 11.5 Startup and switching

| Step | Time |
|---|---|
| First ever start (downloads 6.7 GB) | bandwidth dependent; the five fine-tunes downloaded in 76 to 184 s each on the development connection |
| SDXL load from local cache at start | 34 to 59 s (disk cache dependent) |
| Z-Image Turbo first load (fp32 to bf16 conversion plus writing the 12 GB cache) | about 250 s |
| Z-Image Turbo load from models/cache/ | about 130 s (8-bit quantisation of the text encoder dominates) |
| Qwen-Image first load (quantising 41 GB to NF4 and writing the 11 GB cache) | about 260 s; needs most of the 16 GB system RAM |
| Qwen-Image load from models/cache/ | about 90 s |
| Model switch while running (SDXL family) | 26 to 27 s |
| Upscaler load | under 1 s |
| Admin panel refresh | about 3 s first, 0.7 s after |

## 11.6 Tuning levers

- Model: Z-Image Turbo gives a finished photoreal image in 9 s; DreamShaper Turbo gives a rougher draft in 2 s. For SDXL-specific looks or LoRAs, regenerate the chosen seed with Juggernaut or RealVis.
- Steps: the dominant cost for every model. `quick` halves them; `highly detailed` adds a third.
- Count: images in a batch are generated sequentially, so time scales linearly.
- Upscale later: the Upscale bar costs 3 to 9 s per image and avoids regenerating.
- Cards with less VRAM: generate.py enables CPU offload below 12 GB, which roughly doubles inference time but keeps the pipeline working.